

Question 1: Given the declaration `int matrix [2][3] = {{1, 4, 2}, {3, 6, 8} };`, what is the exact output of `printf ("%d", matrix [1][2]);`?

- A. 2
- B. 8
- C. 3
- D. 6

Question 2: Which memory area is specifically used for “Dynamic Allocation During Execution”?

- A. Stack
- B. Read-only Memory
- C. Heap
- D. CPU register

Question 3: If `int myNumbers [4] = {25, 50, 75, 100};` and `int *p = myNumbers;` what value does `*(p + 3)` evaluates to?

- A. 100
- B. 75
- C. 28
- D. Error

Question 4: Why does `scanf()` fails to read an input example of “Johnny Blaze!” entirely?

- A. It stops at the first whitespace character
- B. It requires a `&` symbol for string
- C. It only accepts numerical values.
- D. The array size is automatically set to 5.

Question 5: What is the correct “Scanset” format in `scanf()` to read a full line including spaces?

- A. `%s`
- B. `%[^\n]`
- C. `%line`
- D. `%All`

Question 6: What is the output of the following string code snippet?

```
char msg[] = "Lab";  
printf("%c", msg[1]);
```

- A. L
- B. a
- C. b
- D. `\0`

Question 7: What does the format specifier `%p` display when used in `printf`?

- A. The value of a variable
- B. The percentage of memory used
- C. A memory address (in hexadecimal)
- D. A protected variable

Question 8: How many bytes does a single `int` variable typically occupy on a modern 64-bit system?

- A. 1 byte
- B. 2 bytes
- C. 4 bytes
- D. 8 bytes

Question 9: In the declaration `char* ptr_arr[3]`; what does each element of the array hold?

- A. A memory address (pointer)
- B. A single character
- C. A whole integer
- D. A function name

Question 10: Which function from `<stdio.h>` is used to read a full line of text safely?

- A. `gets()`
- B. `scanf_s()`
- C. `fgets()`
- D. `fgetc()`

Question 11: `Calloc()` does what?

- A. No initialization
- B. Initializes to zero
- C. Frees memory
- D. Resizes memory

Question 12: If a variable is declared inside a function (e.g., `void myFunction() { int x = 10; }`), can it be accessed or modified directly within the `main()` function??

- A. Yes, all variables in C are global by default.
- B. Yes, if you use the `extern` keyword
- C. No, because the variable has "Local Scope" and only exists within that function.
- D. No, unless it is a pointer variable.